



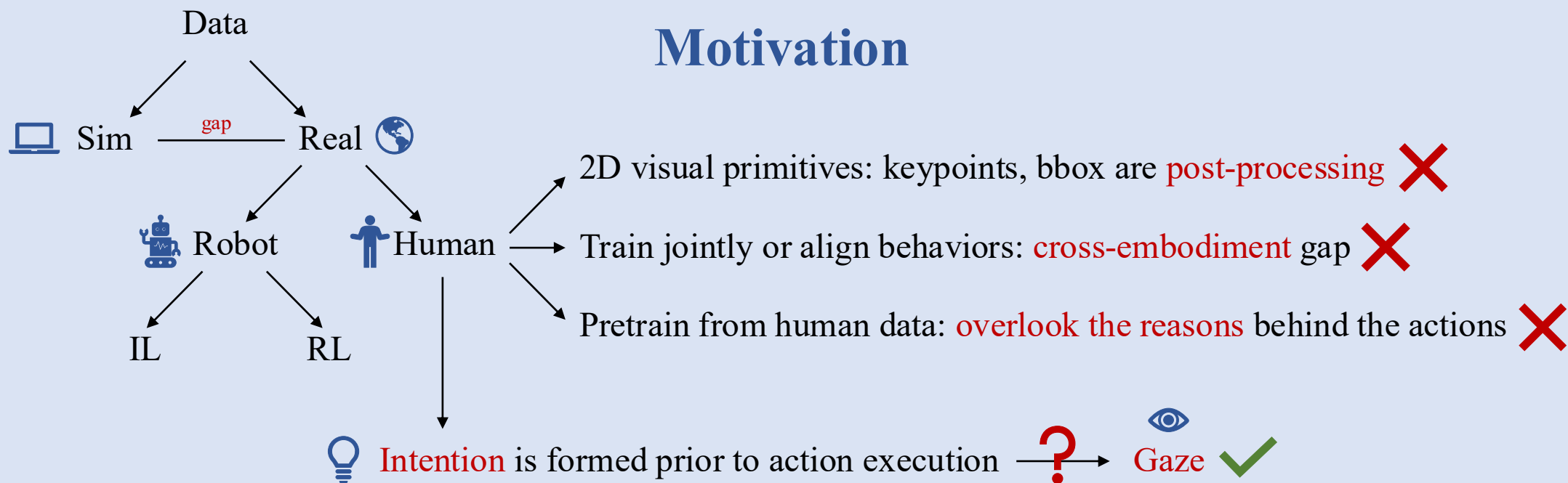
西北工业大学
NORTHWESTERN POLYTECHNICAL UNIVERSITY

VLA **→** **VLIA**

Zhengwei Yang

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Motivation

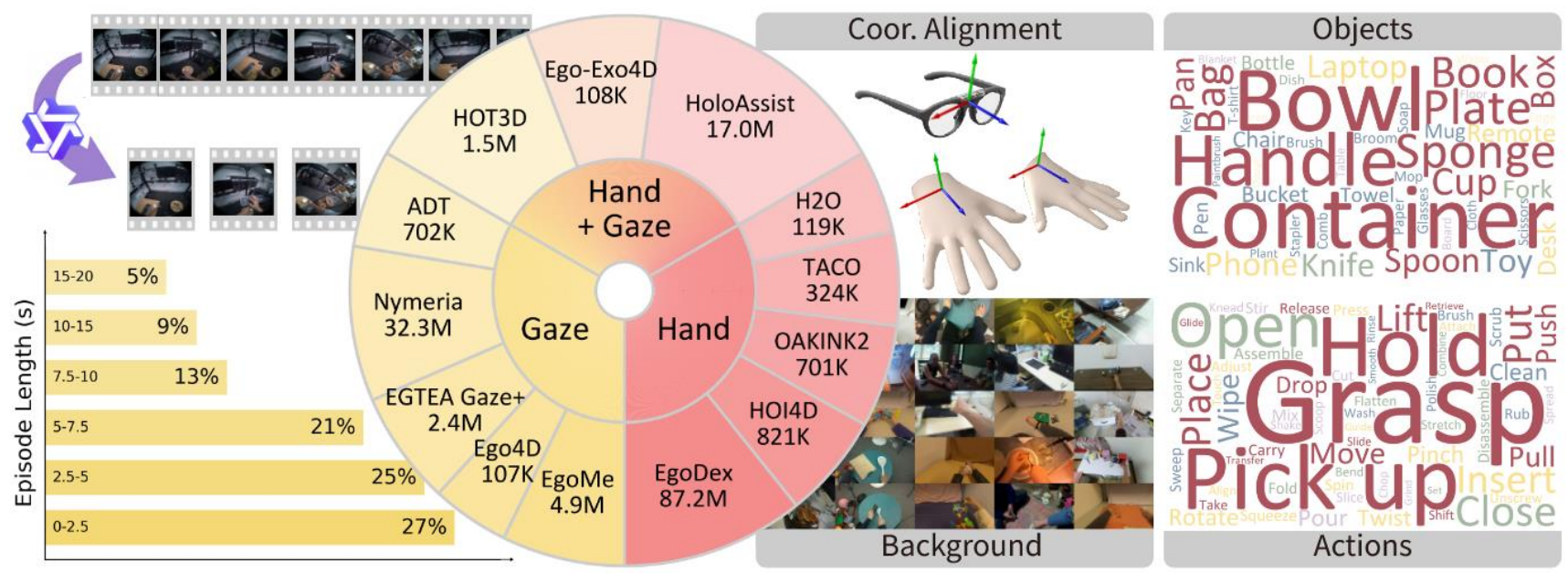


Challenge

How to model **intention** to capture the causal structure of manipulation behavior ?

How to build a training and inference framework with **intention**?

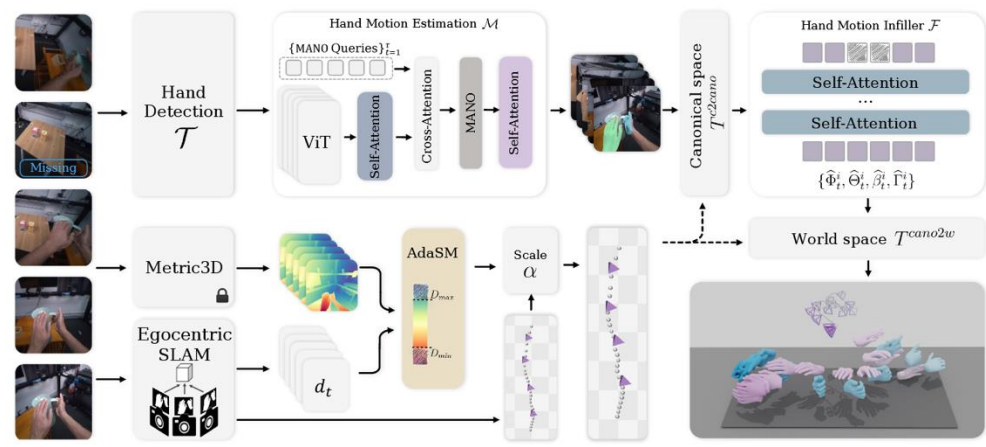
Dataset
with
intention



Large-scale egocentric dataset for pre-training



Pupil Neon glasses



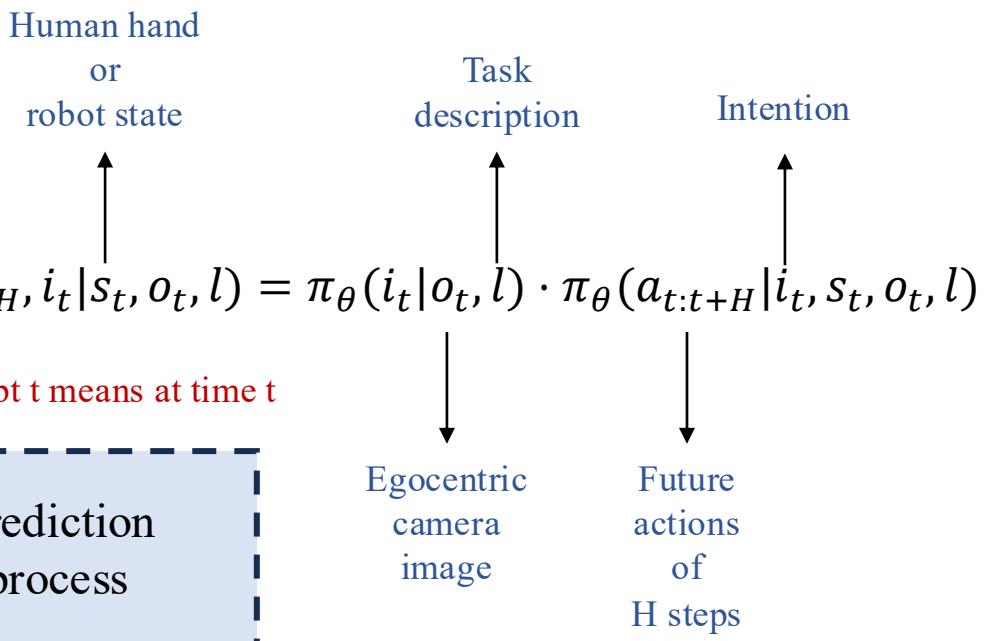
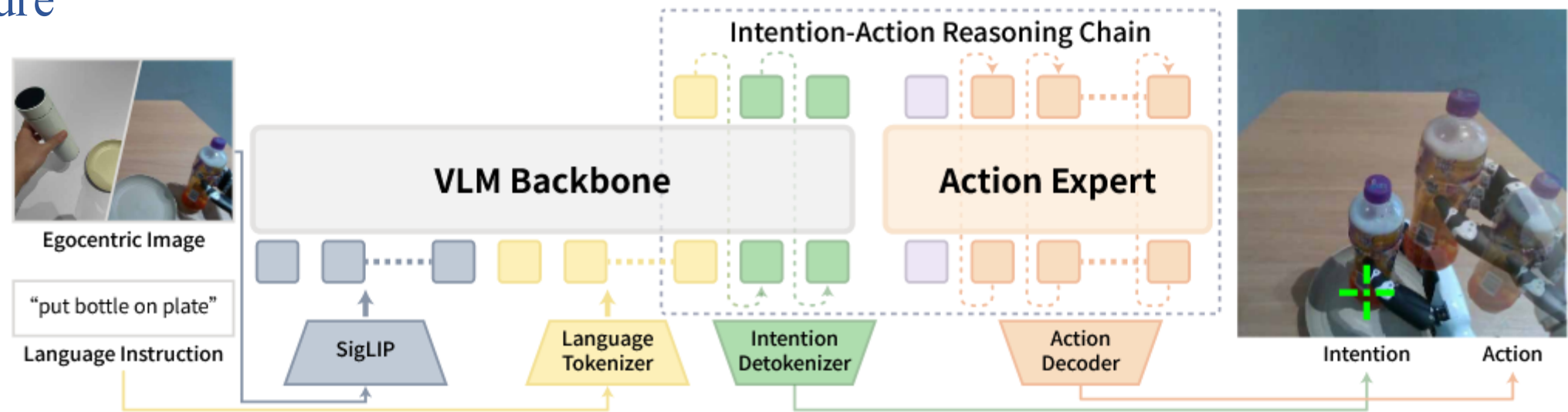
HaWoR

with temporal smoothing and noise filtering



Additional human
demonstrations
for post-training

Architecture



*The subscript t means at time t

perception → intention inference → action generation

$$\mathcal{L}_{\text{intent}} = \mathbb{E} \left[- \sum_{n=1}^{N_{\text{intent}}} \log \pi_{\theta}(i_t^n | o_t, l, i_t^{<n}) \right]$$

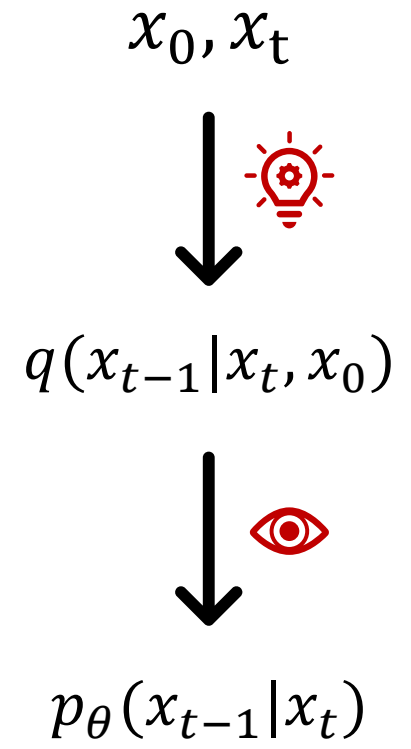
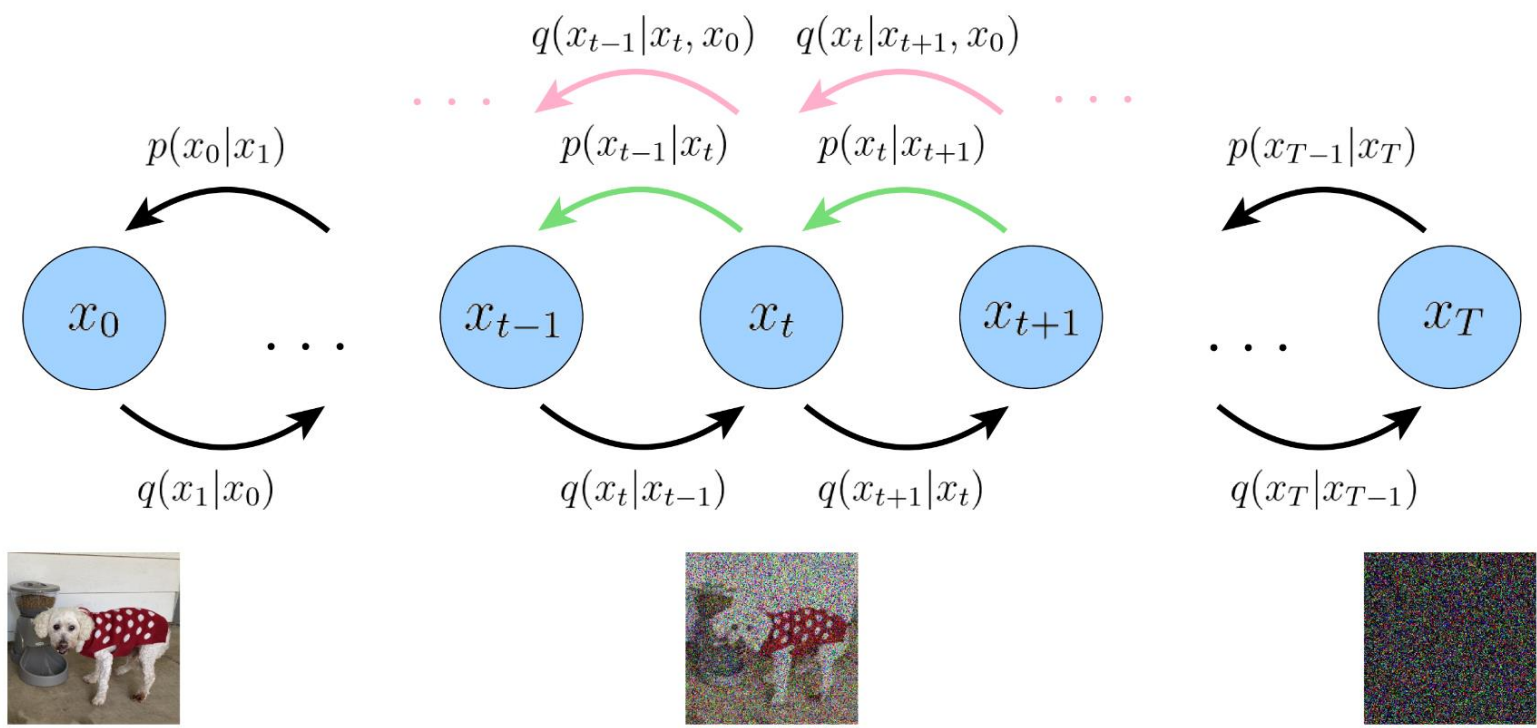
$$\mathcal{L}_{\text{action}} = \mathbb{E} \left[\left\| \pi_{\theta}(a_{t:t+H}^{\tau}, \tau, c) - (a_{t:t+H} - \epsilon) \right\|_2^2 \right]$$

$c = \{i_t, s_t, o_t, l\}$

$a_{t:t+H}^{\tau} = \tau a_{t:t+H} + (1 - \tau) \epsilon$

$$\mathcal{L} = \lambda_{\text{action}} \mathcal{L}_{\text{action}} + \lambda_{\text{intent}} \mathcal{L}_{\text{intent}} \quad * \lambda_{\text{action}} = 1, \lambda_{\text{intent}} = 0.1$$

Loss function



$$q(x_{t-1}|x_t, x_0) = \frac{q(x_{t-1}, x_t, x_0)}{q(x_t, x_0)} = \frac{q(x_t | x_{t-1}, x_0)q(x_{t-1} | x_0)q(x_0)}{q(x_t | x_0)q(x_0)} = \frac{q(x_t | x_{t-1}, x_0)q(x_{t-1} | x_0)}{q(x_t | x_0)}$$

π

π_0 : A Vision-Language-Action **Flow** Model for General Robot Control

Training

Training example

$$\mathbf{A}_t = a_{t:t+H}, c = \{i_t, s_t, o_t, l\}$$



Random sampling
 $\epsilon \sim \mathcal{N}(0, \mathbf{I}), \tau \sim \mathcal{U}(0, 1)$

Computing noisy actions

$$\mathbf{A}_t^\tau = \tau \mathbf{A}_t + (1 - \tau) \epsilon$$



Computing
denoising vector



Network
outputs

$$\mathbf{A}_t = \mathbf{A}_t^\tau + (1 - \tau)(\mathbf{A}_t - \epsilon) \longleftrightarrow \pi_\theta(\mathbf{A}_t, \tau, c)$$

Matching



$$\mathcal{L}_{\text{action}} = \mathbb{E} \left[\left\| \pi_\theta(\mathbf{A}_t, \tau, c) - (\mathbf{A}_t - \epsilon) \right\|_2^2 \right]$$

Inference

$$\mathbf{A}_t^\tau = \tau \mathbf{A}_t + (1 - \tau) \epsilon$$

$$\mathbf{A}_t^0 = \epsilon \sim \mathcal{N}(0, \mathbf{I}) \longrightarrow \mathbf{A}_t^1 = \mathbf{A}_t$$

$\tau: 0 \rightarrow 1$
Denoising



Euler
integration

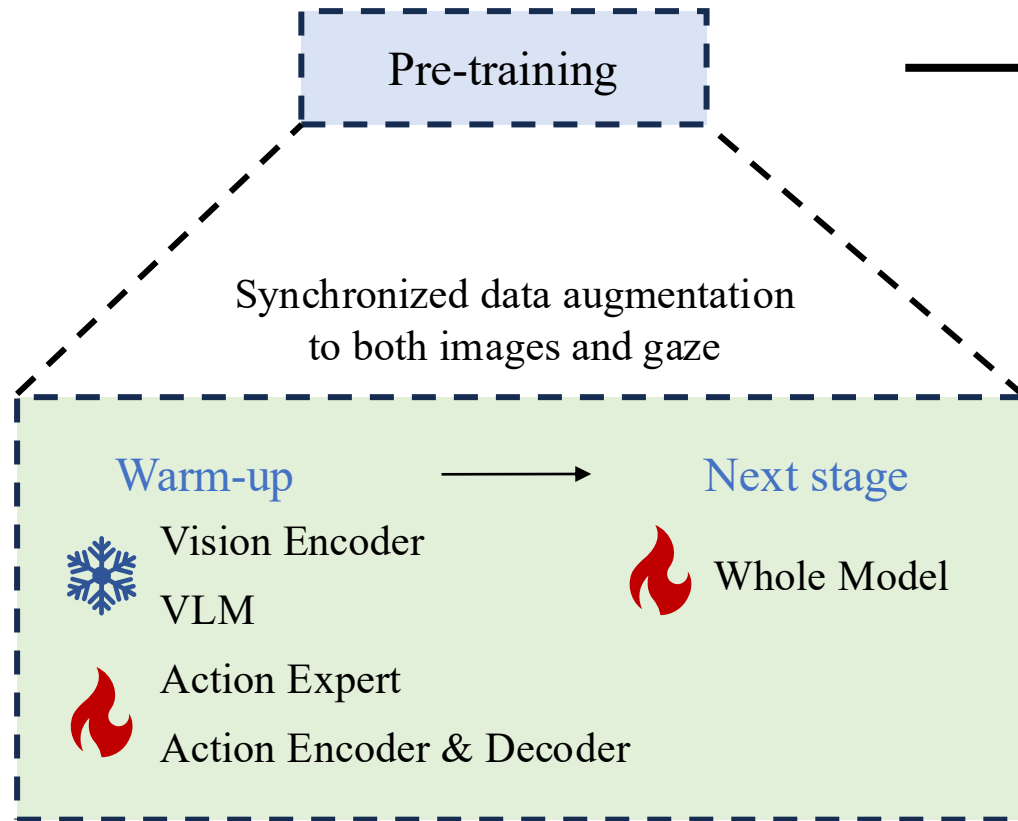
$$\mathbf{A}_t^{\tau+\delta} = \mathbf{A}_t^\tau + \delta \pi_\theta(\mathbf{A}_t, \tau, c)$$

Discussion on choice of δ :

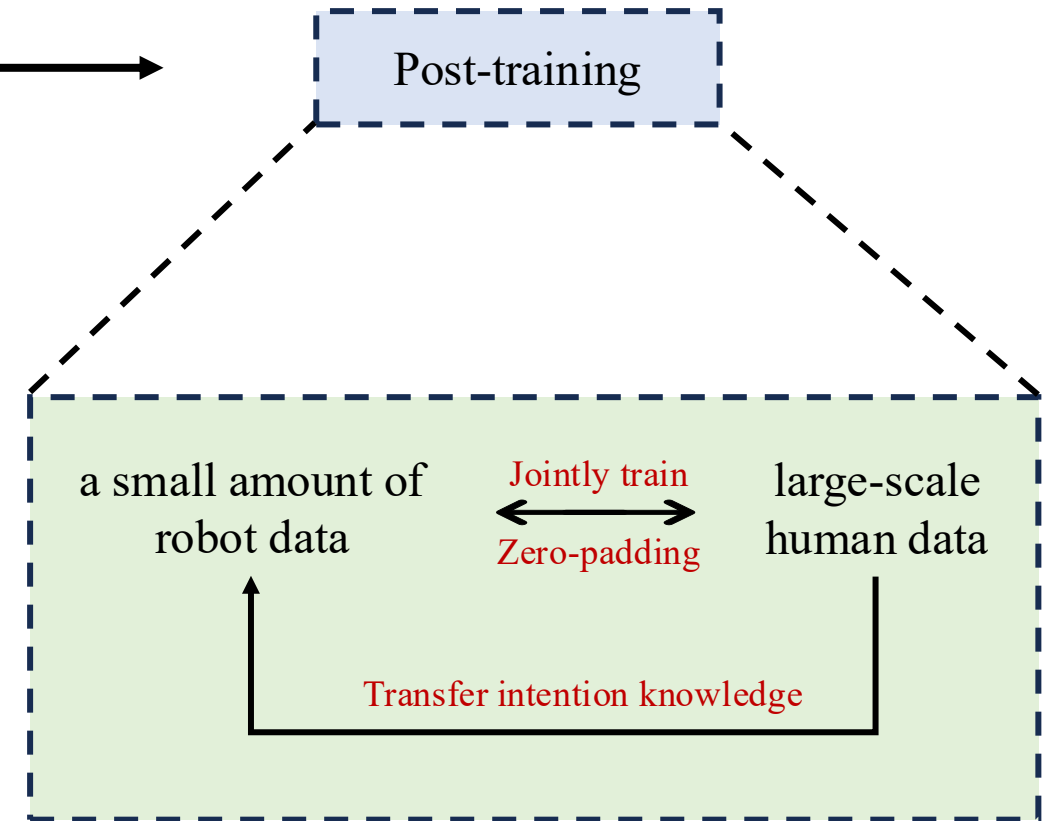
The Euler integration rule is essentially based on **first-order Taylor expansion**, with a local truncation error of $\mathcal{O}(\delta^2)$

large $\delta \rightarrow$ overshooting and increased integration error
small $\delta \rightarrow$ high computational cost

Training



- Stabilize early training
- Prevent representation collapse
- Retain the capabilities of the foundation model

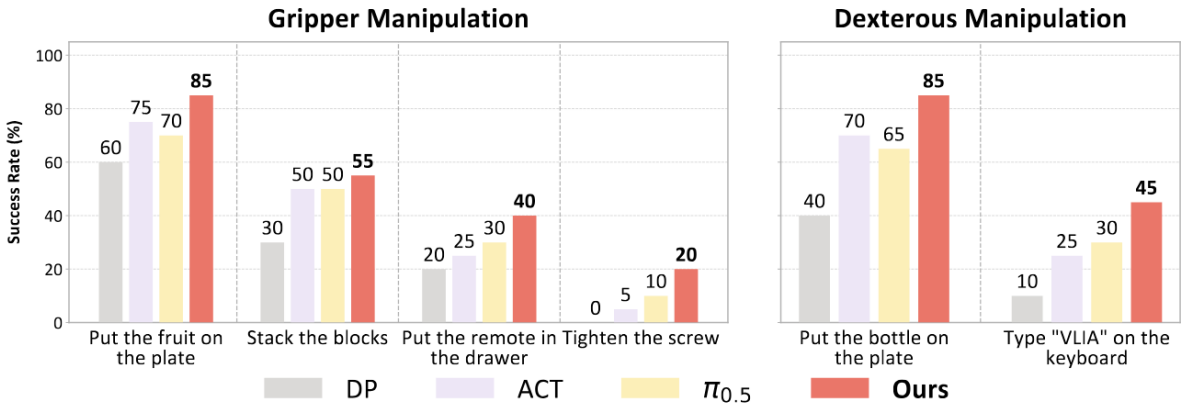


- Improve task performance and generalization

Results

Task \ Method	ID					OOD-Distractors					OOD-Lighting				
	lfa	dp	hrdt	$\pi_{0.5}$	ours	lfa	dp	hrdt	$\pi_{0.5}$	ours	lfa	dp	hrdt	$\pi_{0.5}$	ours
cube transfer	87	75	89	94	100	12	9	28	25	35	0	0	12	32	36
hook package	23	7	13	20	32	8	3	0	11	14	0	0	0	10	19
peg insertion	10	2	17	15	18	7	0	2	6	9	0	0	0	7	16
pour test tube	41	23	24	34	39	11	7	14	24	28	0	0	3	25	22
slot insertion	43	32	42	54	60	23	12	19	47	56	0	0	11	44	50
thread needle	56	30	46	33	43	21	9	23	19	23	0	0	12	20	21
average	43	28	39	41	49	14	7	14	22	28	0	0	6	23	27

Simulation

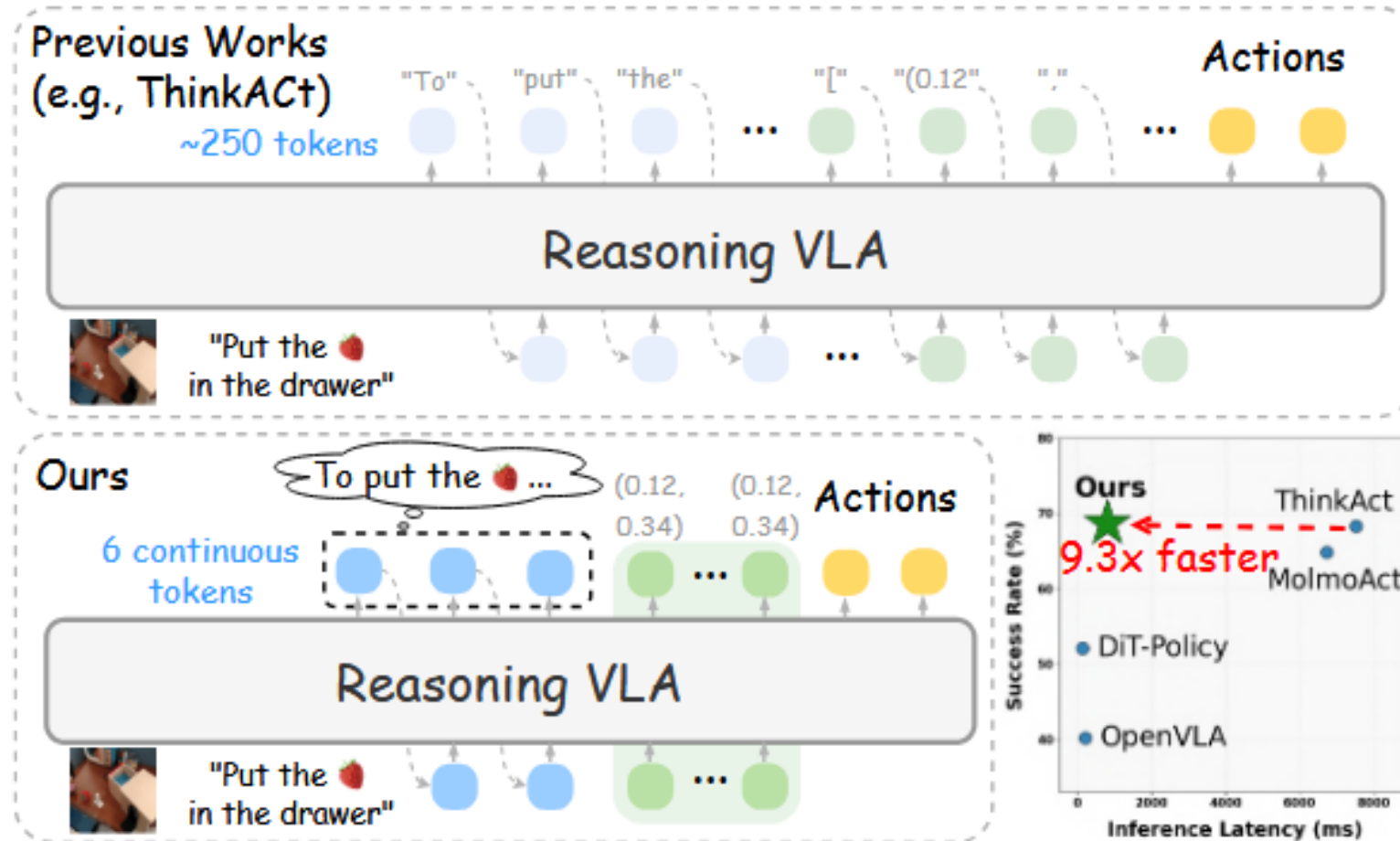


Real-world

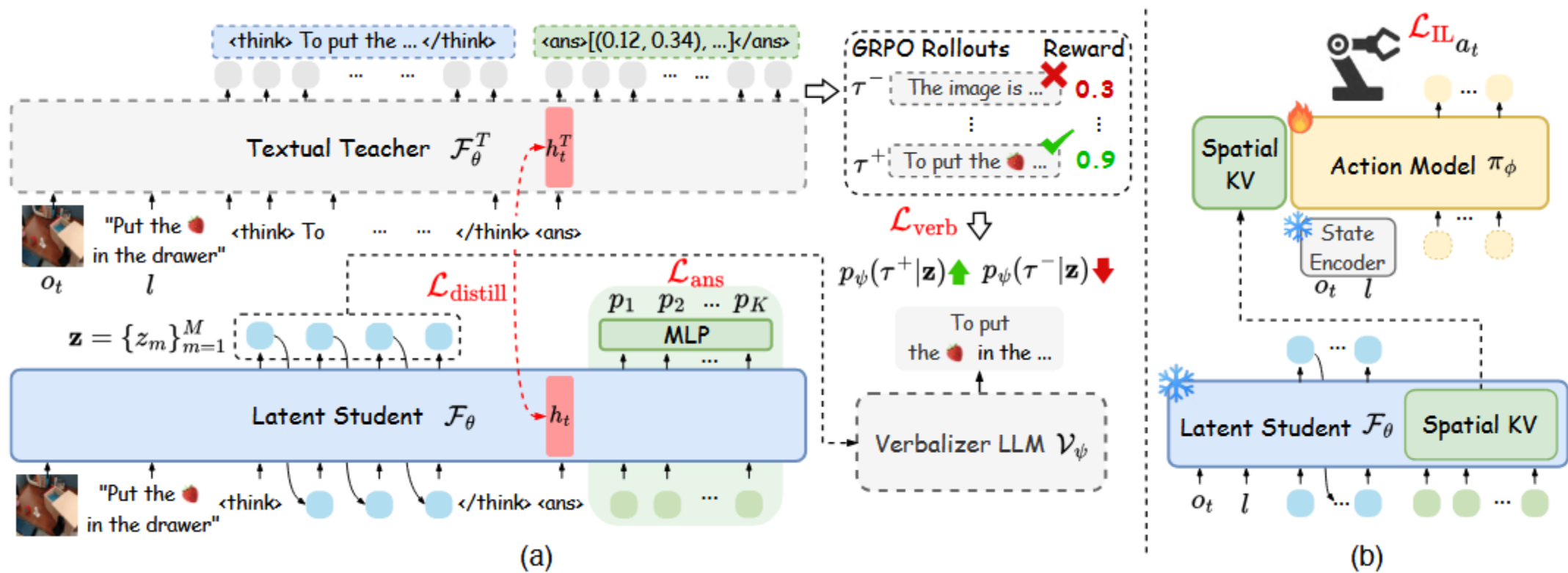
Ablation Study

Method		ID	OOD-Position	OOD-Object	OOD-Scene
ACT		16/20	1/10	4/10	0/10
DP		13/20	2/10	3/10	0/10
$\pi_{0.5}$		17/20	3/10	4/10	3/10
•	ours w/o CoT	16/20	5/10	6/10	5/10
•	• ours (robot only)	14/20	2/10	4/10	0/10
•	ours (robot+human finetune)	13/20	1/10	2/10	0/10
•	ours (robot+human pretrain)	17/20	3/10	5/10	2/10
• • •	ours	19/20	6/10	8/10	6/10

- Learn intention from human data and successfully transfer it to robotic execution
- The causal reasoning paradigm that infers intention before action leads to improved manipulation performance
- Learning intention from human data substantially enhances generalization across variations in object positions, object categories, and scenes
- human data play an important role in both the pretraining and finetuning stages



Comparison of ThinkAct & Fast-ThinkAct



Overview of Fast-ThinkAct



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Thanks for listening!

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